Class 33: Calculating Allowed Energy Bands and Forbidden Band Gaps

In this class we wish to look at the origin of band gaps in an analytical manner. The approach for this is credited to Kramer and is shown in the Figure 33.1 below.

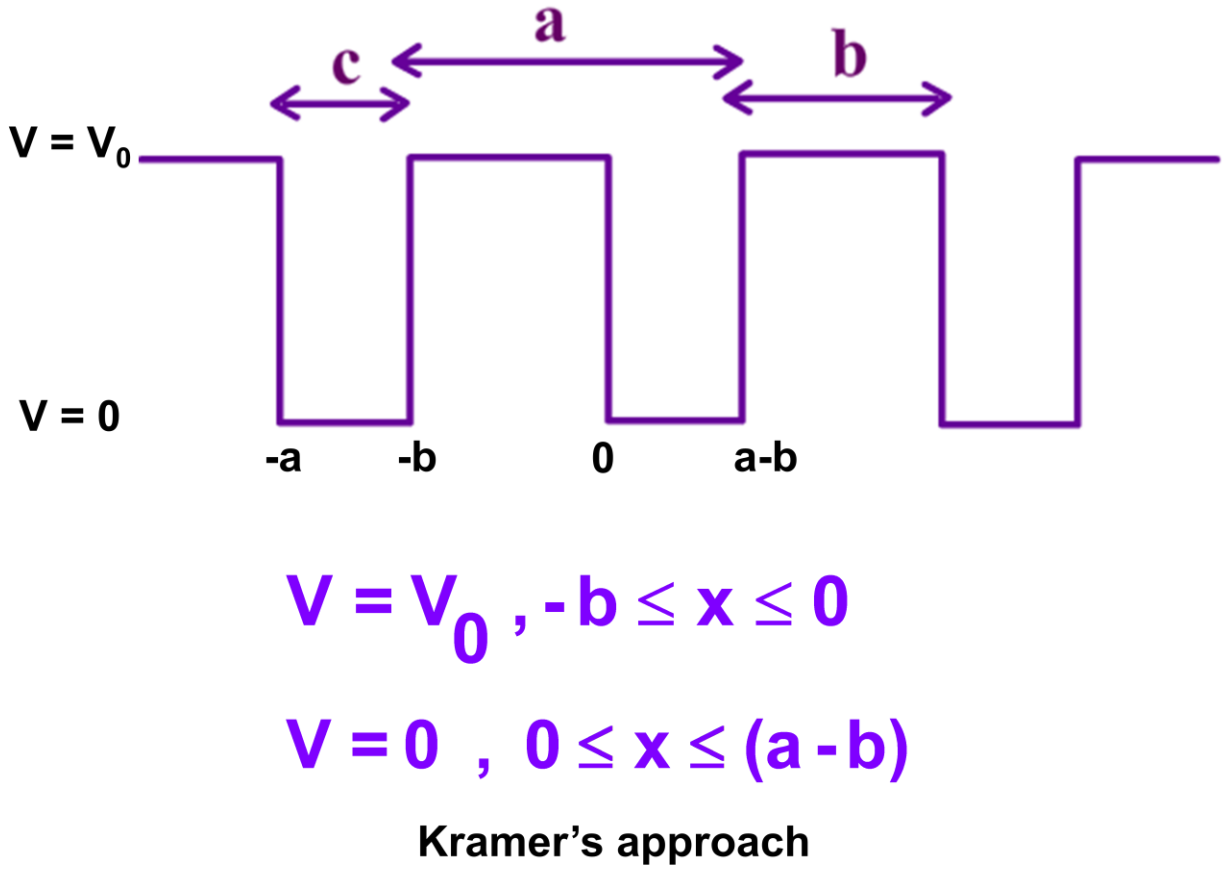


Figure 33.1: The dimensions of the potential wells, and the potentials involved, for the prupose of determining the band gaps corresponding to a solid.

The Schrödinger wave equation in one dimension is as follows:

$$\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}+V(x)\right)\psi(x)=E\psi(x)$$

This equation is solved for regions where $V=0$ and $V=V_0$, and the results are compared. In the equations that follow, it is important to note the subscripts to keep track of the potential in the region where the calculation is being done. As indicated above, typically at the boundary, on one side $V=0$ and on the other side $V=V_0$. So calculations in these two regions will be compared with each other in many of the equations that follow.

When $V=0$,

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2}=E\psi(x)$$

Or

$$\frac{d^2\psi(x)}{dx^2} = \frac{-2mE}{\hbar^2}\psi(x)$$

Which can be solved by examination and we get solution of the form:

$$\psi(x) = A\exp(i\beta x) + B\exp(-i\beta x)$$

Where

$$\beta = \frac{\sqrt{2mE}}{\hbar}$$

And A, and B are constants that need to be determined

Similarly, when $V = V_0$,

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V_0\psi(x) = E\psi(x)$$

Or

$$\frac{d^2\psi(x)}{dx^2} = \frac{2m(V_0 - E)}{\hbar^2}\psi(x)$$

Which can be solved by examination and we get solution of the form:

$$\psi(x) = C\exp(\alpha x) + D\exp(-\alpha x)$$

Where

$$\alpha = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$$

And C, and D are constants that need to be determined

For the solutions to be reasonable, we require $\psi(x)$ to be continuous, therefore at $x = 0$, we require

$$\psi_0(0) = \psi_V(0)$$

And

$$\frac{d\psi_0(0)}{dx} = \frac{d\psi_V(0)}{dx}$$

Periodicity has been expressed mathematically by Bloch as follows:

$$\psi(x + a) = \psi(x)\exp(ika)$$

Therefore, comparing the two positions $x = -b$, and $x = a - b$, which are related by the periodicity of the lattice, we have:

$$\psi_0(a-b) = \psi_V(-b)\exp(ika)$$

or

$$\psi_V(-b) = \exp(-ika)\,\psi_0(a-b)$$

And

$$\frac{d\psi_V(-b)}{dx} = \exp(-ika)\frac{d\psi_0(a-b)}{dx}$$

Therefore, we have the following four equations:

$$\psi_0(0) = \psi_V(0)$$

$$\frac{d\psi_0(0)}{dx} = \frac{d\psi_V(0)}{dx}$$

$$\psi_V(-b) = \exp(-ika)\,\psi_0(a-b)$$

$$\frac{d\psi_V(-b)}{dx} = \exp(-ika)\frac{d\psi_0(a-b)}{dx}$$

Based on the solutions at $V = 0$ and $V = V_0$, the above four equations result in the following:

$$A + B = C + D$$

$$i\beta(A - B) = \alpha(C - D)$$

$$\exp(-ika)\,[A\exp(i\beta(a-b)) + B\exp(-i\beta(a-b))] = C\exp(-\alpha b) + D\exp(\alpha b)$$

$$i\beta[\exp(-ika)][A\exp(i\beta(a-b)) - B\exp(-i\beta(a-b))] = \alpha C\exp(-\alpha b) - \alpha D\exp(\alpha b)$$

Therefore we have four equations in A, B, C and D.
These can be solved if the determinant corresponding to their coefficients is equated to zero.
i.e.:

$$\begin{vmatrix} 1 & 1 & -1 & -1 \\ i\beta & -i\beta & -\alpha & \alpha \\ \exp(-ika)\exp i\beta(a-b) & \exp(-ika)\exp[-i\beta(a-b)] & -\exp(-\alpha b) & -\exp(\alpha b) \\ i\beta\exp(-ika)\exp i\beta(a-b) & -i\beta\exp(-ika)\exp[-i\beta(a-b)] & -\alpha\exp(-\alpha b) & \alpha\exp(\alpha b) \end{vmatrix}$$

The determinant above should evaluate or equate to zero

Since

$$\cosh x = \frac{1}{2}\,(e^{x} + e^{-x})$$

And

$$\sinh x = \frac{1}{2}\,(e^{x} - e^{-x})$$

We can rewrite the above determinant and designate the columns as C1, C2, C3, and C4.

C1	C2	C3	C4
1	1	-1	-1
$i\beta$	$-i\beta$	$-\alpha$	α
$\exp(-ika)\exp i\beta(a-b)$	$\exp(-ika)\exp[-i\beta(a-b)]$	$-\exp(-\alpha b)$	$-\exp(\alpha b)$
$i\beta\exp(-ika)\exp i\beta(a-b)$	$-i\beta\exp(-ika)\exp[-i\beta(a-b)]$	$-\alpha\exp(-\alpha b)$	$\alpha\exp(\alpha b)$

If we designate the columns of the determinant as C1, C2, C3 and C4 and first carry out the operations C1 – C2 , C4 – C3; and follow that with the operation C3+C2, we get

0	1	0	0
$2i\beta$	$-i\beta$	$-\alpha-i\beta$	2α
$2i\exp(-ika)\sin\beta(a-b)$	$\exp(-ika)\exp(-i\beta(a-b))$	$\exp(-ika)\exp(-i\beta(a-b))-\exp(-\alpha b)$	$-2\sinh(\alpha b)$
$2i\beta\exp(-ika)\cos\beta(a-b)$	$-i\beta\exp(-ika)\exp(-i\beta(a-b))$	$-i\beta\exp(-ika)\exp(-i\beta(a-b))-\alpha\exp(-\alpha b)$	$+2\alpha\cosh\alpha b$

Expanding the determinant using row 1, or obtaining its Laplace expansion, we see that the determinant evaluates to $A_{12}C_{12}$ where C_{12} is the cofactor of A_{12}

Since A_{12} is 1, we can simplify our problem to evaluating C_{12} which is:

$$(-1)^{1+2}\begin{vmatrix} 2i\beta & -\alpha - i\beta & 2\alpha \\ 2ie^{(-ika)}\sin\beta(a-b) & e^{(-ika)}e^{(-i\beta(a-b))} - e^{(-\alpha b)} & -2\sinh\alpha b \\ 2i\beta\, e^{(-ika)}\cos\beta(a-b) & -i\beta e^{(-ika)}e^{(-i\beta(a-b))} - \alpha e^{(-\alpha b)} & 2\alpha\cosh\alpha b \end{vmatrix}$$

Since the determinant must evaluate to zero, we can ignore the constant $(-1)^3$ as well the multiplicative factor 2 in column 1 as well as in column 3. Therefore the determinant now becomes:

$$\begin{vmatrix} i\beta & -\alpha - i\beta & \alpha \\ ie^{(-ika)}\sin\beta(a-b) & e^{(-ika)}e^{(-i\beta(a-b))} - e^{(-\alpha b)} & -\sinh\alpha b \\ i\beta\, e^{(-ika)}\cos\beta(a-b) & -i\beta e^{(-ika)}e^{(-i\beta(a-b))} - \alpha e^{(-\alpha b)} & \alpha\cosh\alpha b \end{vmatrix}$$

Adding columns 1 and 3 to column 2, we get:

$$\begin{vmatrix} i\beta & 0 & \alpha \\ i\exp(-ika)\sin\beta(a-b) & \exp(-ika)\cos\beta(a-b)-\cosh\alpha b & -\sinh\alpha b \\ i\beta\exp(-ika)\cos\beta(a-b) & -\beta\exp(-ika)\sin\beta(a-b)+\alpha\sinh\alpha b & \alpha\cosh\alpha b \end{vmatrix}$$

Expanding, we obtain:

$$i\beta\left[\exp(-ika)\cos\beta(a-b)\alpha\cosh\alpha b-\alpha\cosh^2\alpha b-\beta\exp(-ika)\sin\beta(a-b)\sinh\alpha b+\alpha\sinh^2\alpha\beta\right]$$
$$+\alpha[-i\beta\exp(-2ika)\sin^2\beta(a-b)+i\exp(-ika)\sin\beta(a-b)\alpha\sinh\alpha b-i\beta\exp(-2ika)$$
$$\cos^2\beta(a-b)+i\beta\exp(-ika)\cos\beta(a-b)\cosh\alpha b]$$

$$i\alpha\beta\exp(-ika)\cos\beta(a-b)\cosh\alpha b-i\alpha\beta-i\beta^2\exp(-ika)\sin\beta(a-b)\sinh\alpha b-i\alpha\beta\exp(-2ika)$$
$$+i\alpha^2\exp(-ika)\sin\beta(a-b)\sinh\alpha b+i\alpha\beta\exp(-ika)\cos\beta(a-b)\cosh\alpha b$$

After removing the common constant $i\exp(-ika)$

$$\alpha\beta\cos\beta(a-b)\cosh\alpha b-\alpha\beta\exp(ika)-\beta^2\sin\beta(a-b)\sinh\alpha b-\alpha\beta\exp(-ika)$$
$$+\alpha^2\sin\beta(a-b)\sinh\alpha b+\alpha\beta\cos\beta(a-b)\cosh\alpha b$$

Since the determinant evaluates to zero, after rearranging to group common terms we get:

$$2\alpha\beta\cos\beta(a-b)\cosh\alpha b + (\alpha^2-\beta^2)\sin\beta(a-b)\sinh\alpha b - \alpha\beta(\exp(ika)+\exp(-ika)) = 0$$

Dividing through out by $2\alpha\beta$, we get

$$\cos\beta(a-b)\cosh\alpha b + \left(\frac{\alpha^2-\beta^2}{2\alpha\beta}\right)\sin\beta(a-b)\sinh\alpha b - \left[\frac{\exp(ika)+\exp(-ika)}{2}\right] = 0$$

Since $\dfrac{e^{ika}+e^{-ika}}{2} = \cos ka$

$$\therefore \frac{\alpha^2-\beta^2}{2\alpha\beta}\sin\beta(a-b)\sinh\alpha b + \cos\beta(a-b)\cosh\alpha b = \cos ka$$

On examining the parameters in the left hand side of the equation, we find

$$\alpha = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$$

And

$$\beta = \frac{\sqrt{2mE}}{\hbar}$$

For both α and β the parameter that can be varied is E, the energy of the electron. The other parameters, a, and b are constant once a crystal structure is chosen and specific atoms are chosen to occupy the lattice sites. So, effectively, once a material is chosen, only E can be varied to obtain various values for the expression on the left hand side of the equation.

While the right hand side of the equation is restricted to assume values in the window ± 1, the left hand side of the equation can evaluate to values outside of this window. This situation is interpreted to mean that those energies for which the left hand side of the equation evaluates to values outside the ± 1 window, are energies that are forbidden for the system.

We therefore see how it is possible to evaluate the allowed energy bands and the forbidden energy band gaps of a system, once specific details of the system are known.